



University
of Glasgow

Lecture 5-part 2: Data Collection

Slides prepared by : Joseph O'Hagan and Marwa Mahmoud

Today's session

- ~~Summarise the videos~~
- ~~Quantitative research methods~~
- ~~Qualitative research methods~~
- Data collection: the 5 Ps
- A note about some data collection terminology
- Data collection methods
- Analysing qualitative data: open, axial, selective coding

Data Collection - 5 Ps

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- **Performance**
- **Preference**
- **Perception**
- **Process**
- **Product**

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- Performance — response time to task, number of errors made, etc
 - Assessing the performance of the human, algorithm, system
 - Answering questions of “how well it performs (with respect to X)”

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- Preference — preference in a comparative task
 - Normally done with a human and you ask “which of these presented systems/approaches do you like the best”

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- Perception — free-form opinions
 - Not just asking for a preference, but asking for opinions from the participants (more broader than preference)
 - Preference + Why

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- Process — actions performed during task
 - The actual actions performed in doing a task (could be human or algorithmic, etc) but it's a detailed record of the actions

Data collection — the 5 Ps

Somewhat artificial but we might think about data collection in terms of:

- **Product — what the participant/system actually created**
 - **The end of the task (process) might produce an object — an artefact — which makes up our product data**

Today's session

- ~~Summarise the videos~~
- ~~Quantitative research methods~~
- ~~Qualitative research methods~~
- ~~Data collection: the 5 Ps~~
- A note about some data collection terminology
- Data collection methods
- Analysing qualitative data: open, axial, selective coding

A note about some terminology

Terminology used throughout user studies

Much of the data collection examples will be based from the perspective of user studies (e.g. HCI studies with participants)

- This is arguably more complex than a systems experiment where the data collection is simply performance metrics of 1000 random instances of the algorithm

So let's revise some key terms you need to be aware of

Terminology used throughout user studies

Quantitative data: numeric data

Qualitative data: non-numeric data

- There are a number of ways in which we could capture these and how you capture them will depend on the experiment being conducted

Terminology used throughout user studies

Close-ended questions: questions designed to be answered with a specific/static response

- Yes/No, on a scale of 1-5, etc

Open-ended questions: designed not to be answerable by a static response

- “Tell me about...”, “Why do you think...”, etc

An example of the terms in action

A user study to evaluate a new input system for text entry in virtual reality

- **What is a quantitative measure?**
- **What is a qualitative measure?**

An example of the terms in action [closed questions]

A user study to evaluate a new input system for text entry in virtual reality

- **Quantitative data — time taken during text entry task, number of errors made, etc**

But also:

- **“How comfortable did you find text entry to be using the system, rate this on a scale of 1-5, (1=very uncomfortable, 5=very comfortable)”**

An example of the terms in action [open questions]

A user study to evaluate a new input system for text entry in virtual reality

- **Qualitative data —**
 - **“How comfortable do you think this new approach is to past approaches you have used”**
 - **“What problems did you encounter while using the new system”**
 - **Etc**

Today's session

- ~~Summarise the videos~~
- ~~Quantitative research methods~~
- ~~Qualitative research methods~~
- ~~Data collection: the 5 Ps~~
- ~~A note about some data collection terminology~~
- Data collection methods
- Analysing qualitative data: open, axial, selective coding

Data collection methods

Data collection — data collection methods (in HCI)

- Questionnaire / Survey
- Interviews
- Task sheet
- Observation
- Automatic Collection
- Artefact / Outcome

Applicable to other domains — HCI ones are a little more complex so we'll discuss these

Data collection — data collection methods (in HCI)

- Questionnaire / Survey
 - Take a variety of possible forms
 - A mixture of quantitative / qualitative questions
 - A mixture of closed/open ended questions
 - Can be a standalone survey study or part of a larger experiment

Data collection — data collection methods (in HCI)

- Interviews

- A face-to-face discussion with a participant
- Structured or semi-structured interview
 - Specific questions that must be asked (can go off topic in semi)
- A mixture of closed/open ended questions
- Can be a standalone study or part of a larger experiment

Data collection — data collection methods (in HCI)

- **Task sheet**
 - **Given a set of tasks to complete**
 - **Get to perform specific actions and get them to record the consequence of these actions, opinions after doing them, etc**

Data collection — data collection methods (in HCI)

- Observation
 - The experimenter observes what is happening
 - Might have asked a participant to use a new system and the experimenter watches the participants actions
 - Record notes of number of mistakes made, number of times certain actions were performed, any interesting behaviors, etc

Data collection — data collection methods (in HCI)

- Automatic collection
 - Where some system collects the information for you
 - E.g. time taken for a task, system metrics
 - E.g. a logging of every action performed — inputs, eye tracking data, physiological signals, etc

Data collection — data collection methods (in HCI)

- Artefact
 - Some object that is produced at the end of the experimental activity

Data collection — data collection methods (in HCI)

	questionnaire	interview	task sheet	observation	automatic collection	Artefact/ outcome
Performance			x	x	x	x
Preference	x	x		x		
Perception	x	x		x		
Process				x	x	x
Product				x		x

Data collection — data collection methods (in HCI)

questionnaire	interview	task sheet	observation	automatic collection	artefact / outcome
easy to administer	not always easy to conduct	easy to administer	obvious thing to do	easy to collect	(maybe) easy to collect
difficult to design	difficult to design	based on task	difficult to gather systematic data	tempting to collect more than is needed	being unambiguous is crucial
typically quick	takes time	takes as long as the task itself	takes as long as the task itself	takes as long as the task itself	takes as long as the task itself
difficult to get detailed qualitative answers	better for detailed qualitative answers	usually collect answers to direct questions	can be qualitative and/or quantitative	time consuming to analyse	qualitative product data
easy to get large numbers of participants	difficult to get participants	availability of participants will depend on recruitment to do the task	availability of participants will depend on recruitment to do the task	availability of participants will depend on recruitment to do the task	availability of participants will depend on recruitment to do the task

Data collection — an example

(gathering feedback on a system)

critterion	Quantitative data	Qualitative data
ease of use	75% of participants thought the system was easy to use	"Rather clumsy"; "Quite usable if you know what you are doing"
quick	90% of participants completed the task in 10seconds	"Got through the task quite quickly – did I get it right?"
intuitive	40% of participants took an incorrect navigation path to the 'scan and print' page	"Not sure where I was"; "Had to search around quite a bit"
fun	85% of participants said they would like to use the system again	"That was great!"; "Can I do it again, please?"
lucrative	72% of participants paid the required £1.50 for the extra features	"Not sure why I had to pay, but I did anyway"; "very cheap"
supports task	No participants made any errors	"The tasks were really easy to complete"

Today's session

- ~~Summarise the videos~~
- ~~Quantitative research methods~~
- ~~Qualitative research methods~~
- ~~Data collection: the 5 Ps~~
- ~~A note about some data collection terminology~~
- ~~Data collection methods~~
- Analysing qualitative data: open, axial, selective coding

Open, Axial, Selective coding

3 step systematic way of analysing qualitative data:

- Open — break textual data down into discrete parts (codes)
- Axial — draw connections between codes
- Selective — select a central category which connects all of the codes

A systematic way of identifying thematic connections in qualitative data

Open coding — breaking the transcript into codes

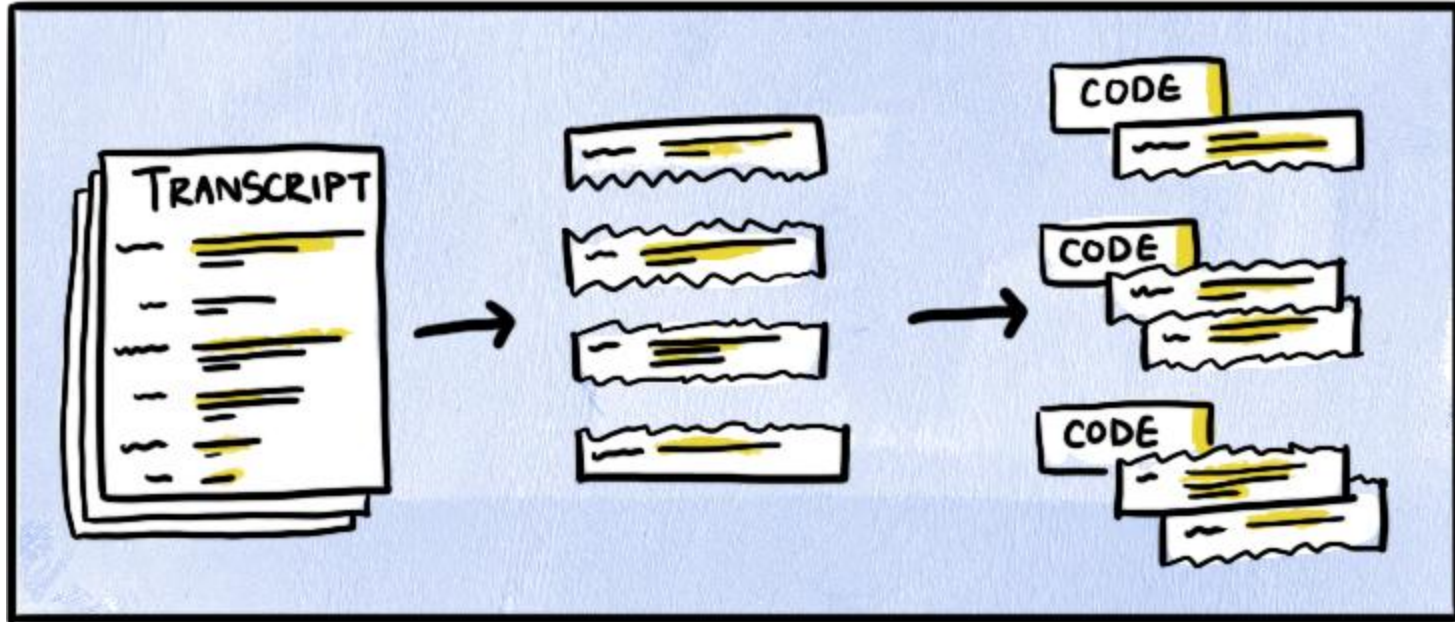


Image taken from: <https://delvetool.com/blog/openaxialselective>

Open coding

In short:

- You read through the data (typically multiple times)
- Then start to create tentative labels for chunks of data (to summarise what is happening)
- Recording participant quotes alongside these tentative labels

Axial coding — draw connections between codes

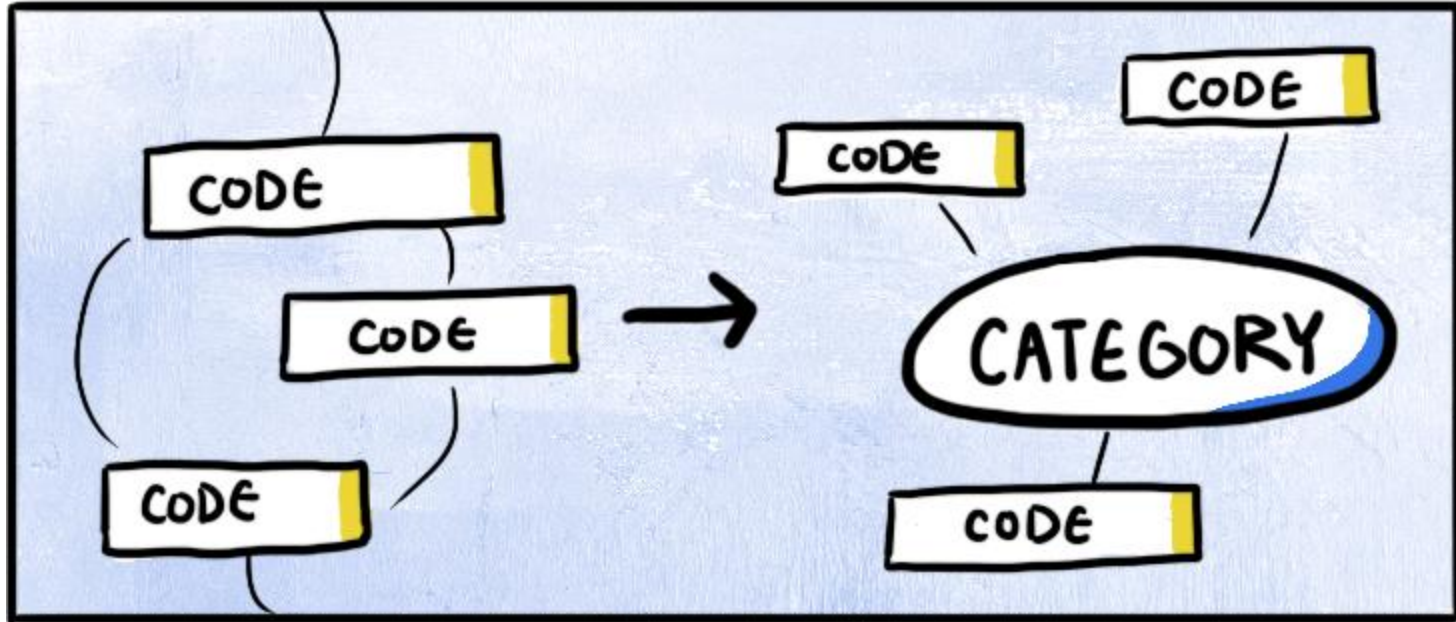


Image taken from: <https://delvetool.com/blog/openaxialselective>

Axial coding

In short:

- You read through your identified codes from the open coding
- Start to map out connections between the codes / grouping similar codes together into categories (thematically grouping codes)

NOTE: categories could be based on an existing code or some more abstract grouping

Selective coding — select a central category

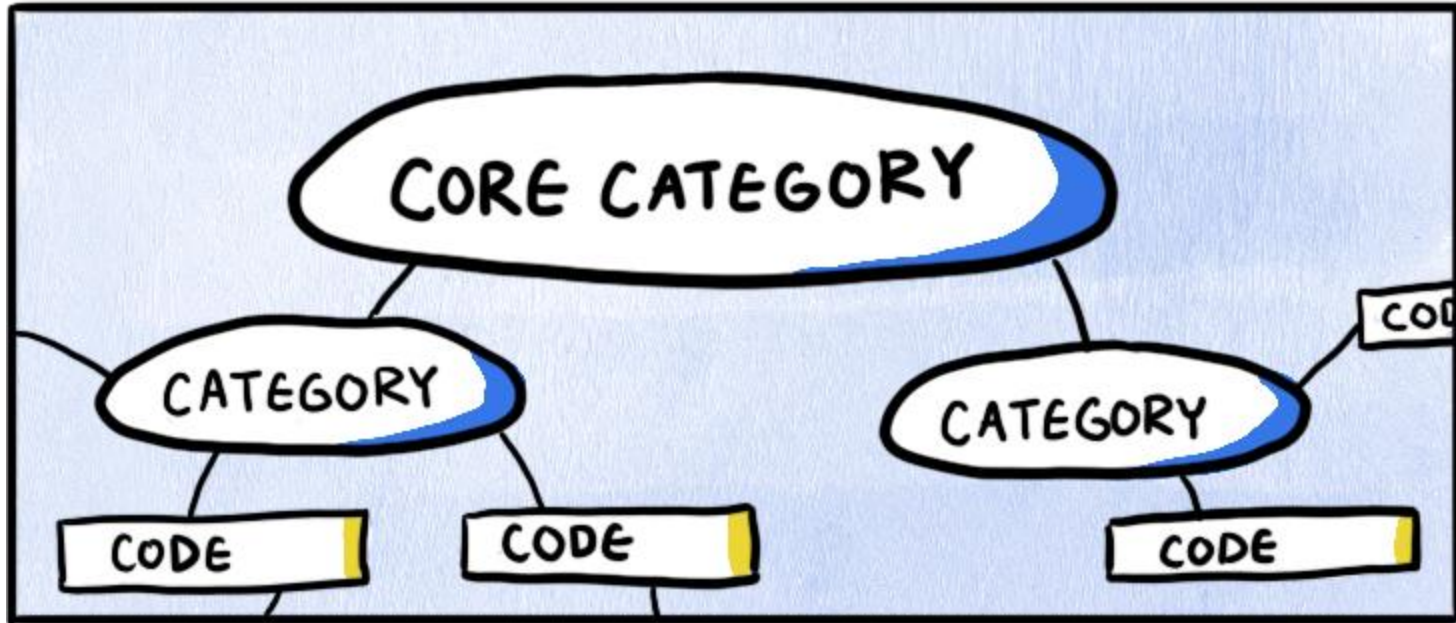


Image taken from: <https://delvetool.com/blog/openaxialselective>

Selective coding

In short:

- You try to identify the highest level category to unify all of the categories and codes
- You might define a new category to do this or you might elevate a category identified at an existing stage

So a 3 step process

Open coding — breaking the transcript into codes

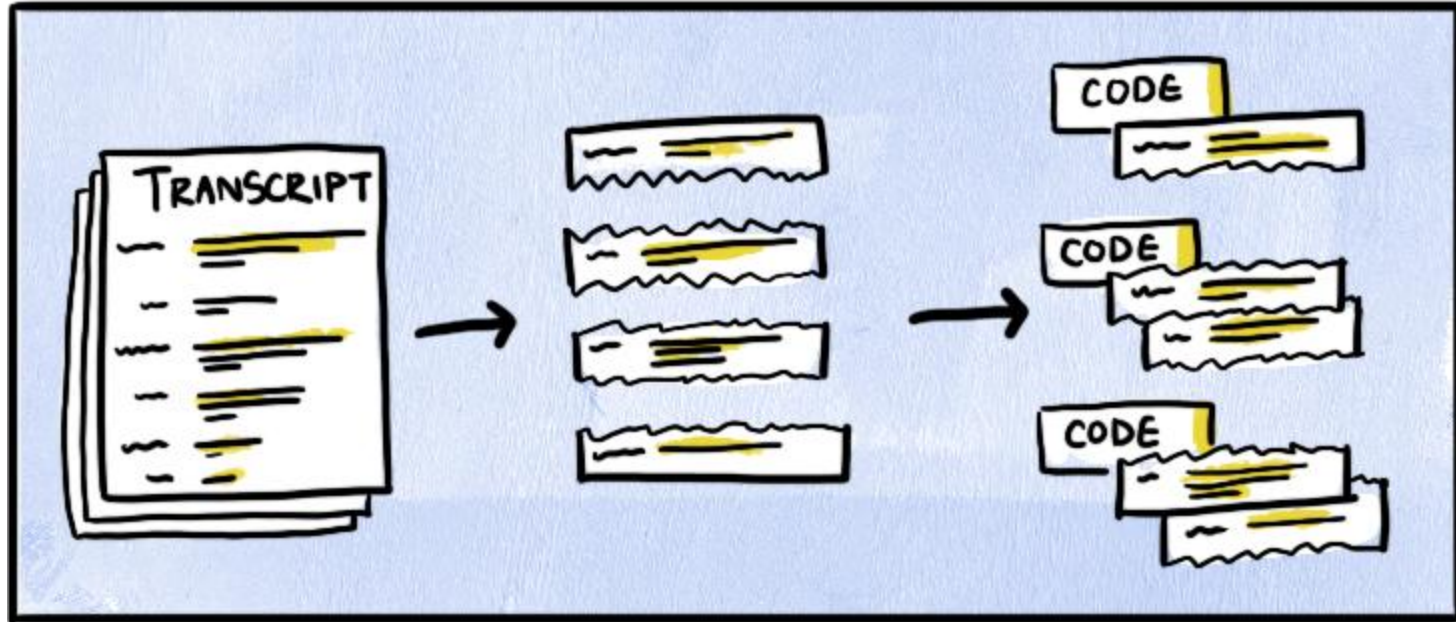


Image taken from: <https://delvetool.com/blog/openaxialselective>

Axial coding — draw connections between codes

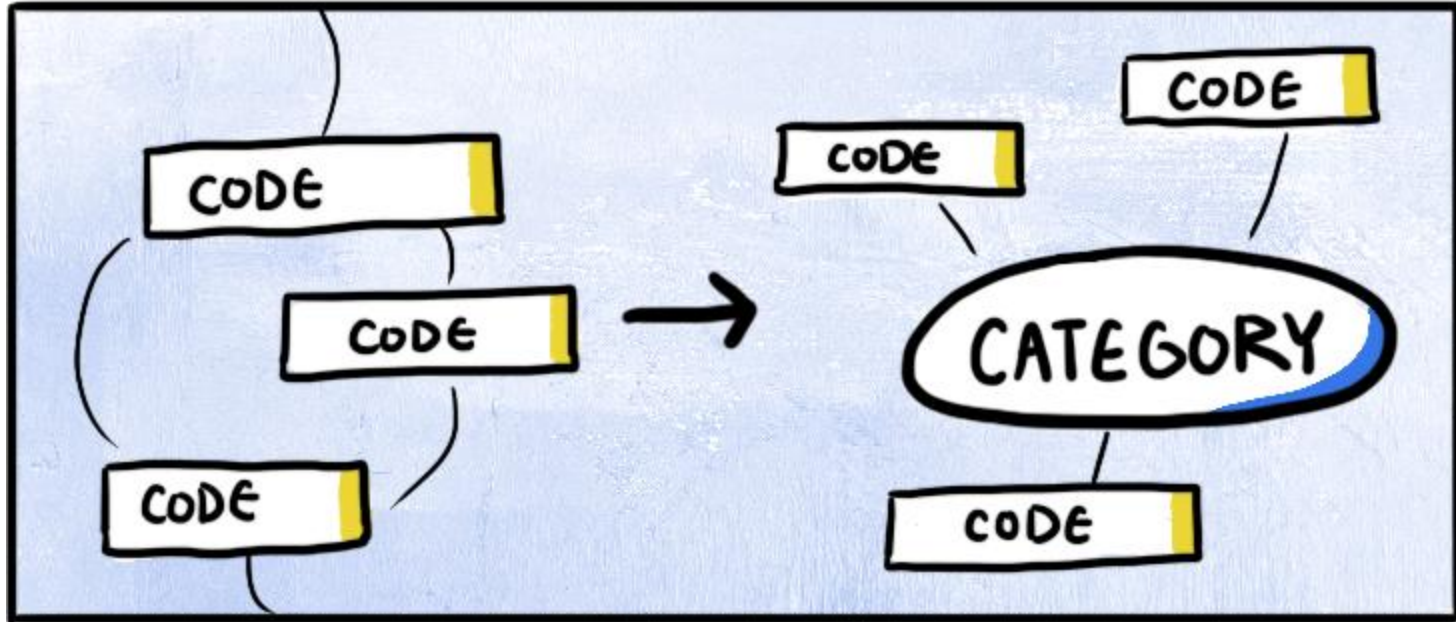


Image taken from: <https://delvetool.com/blog/openaxialselective>

Selective coding — select a central category

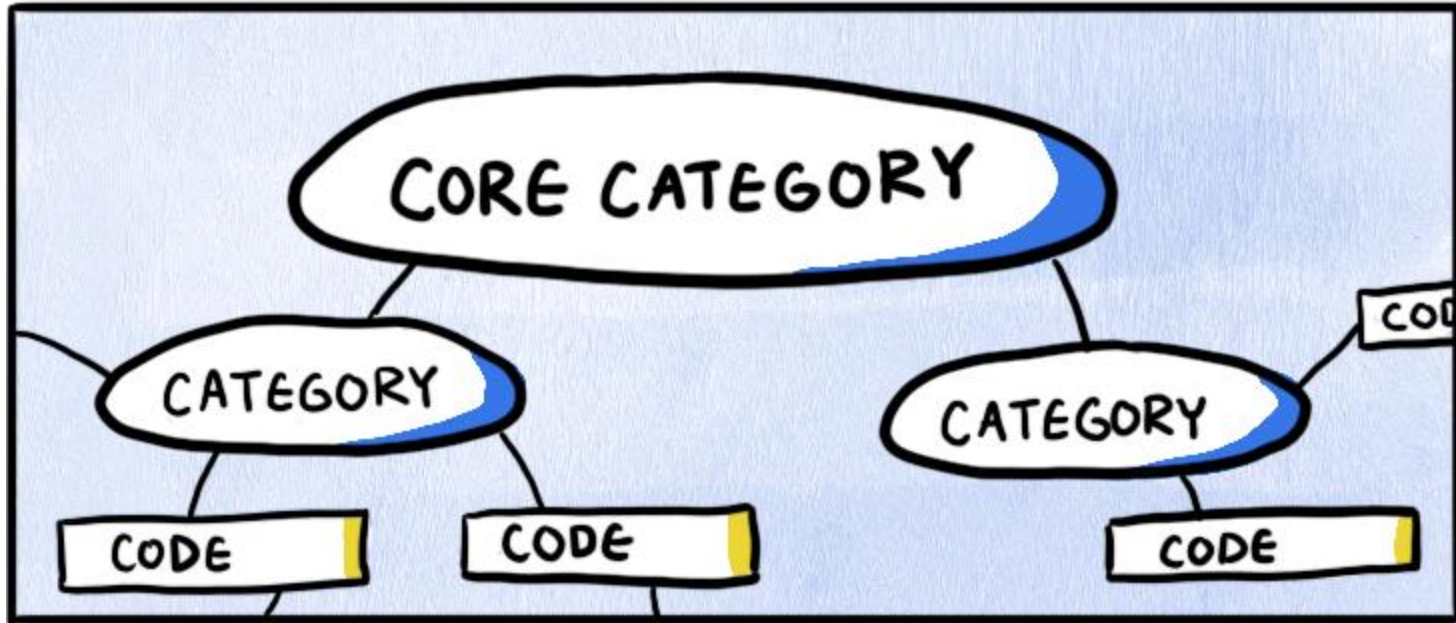


Image taken from: <https://delvetool.com/blog/openaxialselective>

Today's session

- ~~Summarise the videos~~
- ~~Quantitative research methods~~
- ~~Qualitative research methods~~
- ~~Data collection: the 5 Ps~~
- ~~A note about some data collection terminology~~
- ~~Data collection methods~~
- ~~Analysing qualitative data: open, axial, selective coding~~

Summary of today's lecture

- Quantitative and Qualitative research methods
- Data collection terminology and methods
- Some more on ethics
- Mixed methods in computer science